

1-31 In Figure P2-31 find the equivalent resistance between terminals A-B, A-C, A-D, B-C, B-D, and C-D.

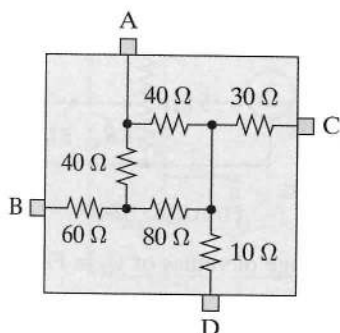


FIGURE P2-31

1-32 Select a value of R_L in Figure P2-32 so that $R_{EQ} = 25 \text{ k}\Omega$. Repeat for $R_{EQ} = 20 \text{ k}\Omega$.

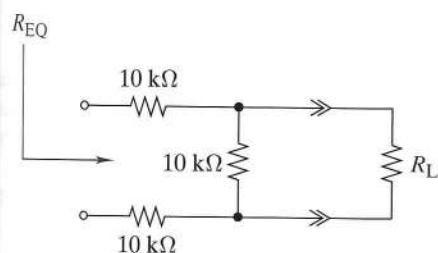


FIGURE P2-32

1-33 Using no more than four 1-kΩ resistors, show how the following equivalent resistors can be constructed: 2 kΩ, 500 Ω, 1.5 kΩ, 333 Ω, 250 Ω, and 400 Ω.

1-34 Find the equivalent practical voltage source at terminals A and B in Figure P2-34.

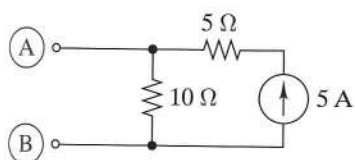


FIGURE P2-34

1-35 In Figure P2-35 the $i-v$ characteristic of network N is $v + 50i = 5 \text{ V}$. Find the equivalent practical current source for the network.

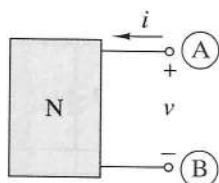


FIGURE P2-35

2-36 Select the value of R_x in Figure P2-36 so that $R_{EQ} = 60 \text{ k}\Omega$.

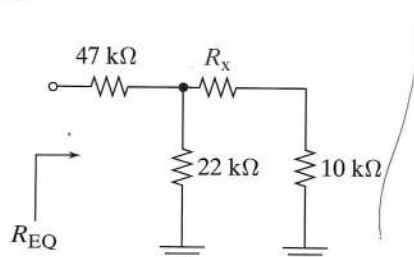


FIGURE P2-36

2-37 Two 10-kΩ potentiometers (a potentiometer is a variable resistor whose value between the two ends is 10 kΩ and between one end and the wiper—the third terminal—can range from 0 Ω to 10 kΩ) are connected as shown in Figure P2-37. What is the range of R_{EQ} ?

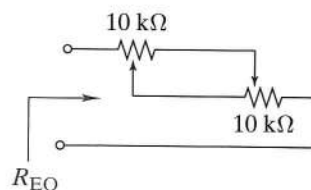


FIGURE P2-37

2-38 Select the value of R in Figure P2-38 so that $R_{AB} = R_L$.

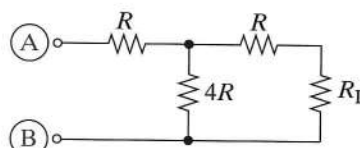


FIGURE P2-38

2-39 What is the range of R_{EQ} in Figure P2-39?

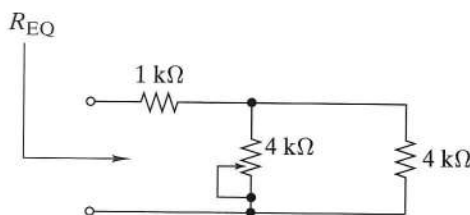


FIGURE P2-39

2-40 Find the equivalent resistance between terminals A and B in Figure P2-40.

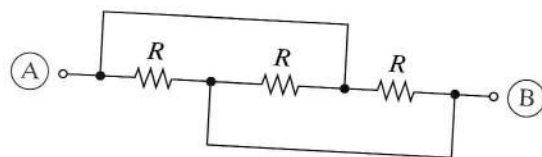


FIGURE P2-40

OBJECTIVE 2-5 VOLTAGE AND CURRENT DIVISION (SECT. 2-5)

- (a) Given a circuit with elements connected in series or parallel, use voltage or current division to find specified voltages or currents.
 (b) Design a voltage or current divider that delivers specified output signals within stated constraints.
 See Examples 2-13, 2-14, 2-16, 2-17, 2-18 and Exercises 2-10, 2-11, 2-12

2-41 Use voltage division in Figure P2-41 to obtain an expression for v_L in terms of R , R_L , and v_S .

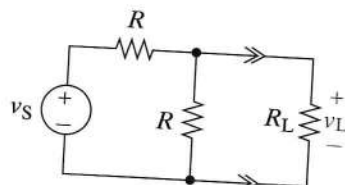


FIGURE P2-41

2-42 Use current division in Figure P2-42 to find v_L if $R = 10\ \Omega$, $R_L = 20\ \Omega$, and $i_S = 2.5\text{ mA}$.

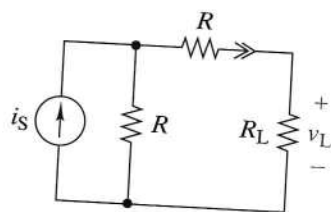


FIGURE P2-42

2-43 Find i_x and i_y in Figure P2-43.

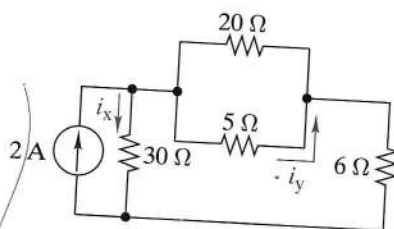


FIGURE P2-43

2-44 The $1\text{-k}\Omega$ load in Figure P2-44 needs 5 V across it to operate correctly. Where should the wiper on the potentiometer be set (R_X) to obtain the desired output voltage?

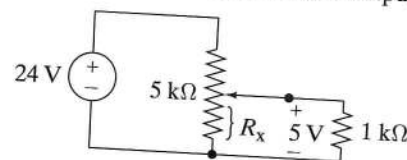


FIGURE P2-44

2-45 Find the range of values of v_O in Figure P2-45.

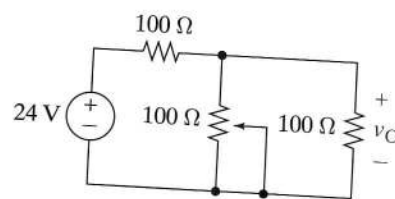


FIGURE P2-45

2-46 Figure P2-46 shows a voltage bridge circuit, that is, two voltage dividers in parallel with a source v_S . One resistor R_X is variable. The goal is often to “balance” the bridge by making $v_X = 0\text{ V}$. Derive an expression for R_X in terms of the other resistors for when the bridge is balanced.

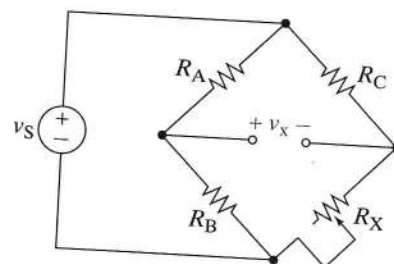


FIGURE P2-46

2-47 Ideally, a voltmeter has infinite internal resistance and can be placed across any device to read the voltage without affecting the result. A particular digital multimeter (DMM), a common laboratory tool, is connected across the circuit shown in Figure P2-47. The expected voltage was 5.73 V . However, the DMM reads 3.81 V . The large, but finite, internal resistance of the DMM was “loading” the circuit and producing a wrong measurement. Find the value of the internal resistance of this DMM.

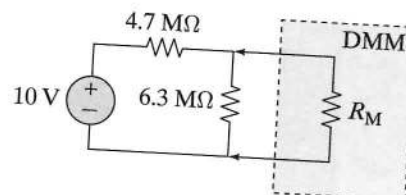


FIGURE P2-47

- 2-48 Select values for R_1 , R_2 , and R_3 in Figure P2-48 so the voltage divider produces the two output voltages shown.

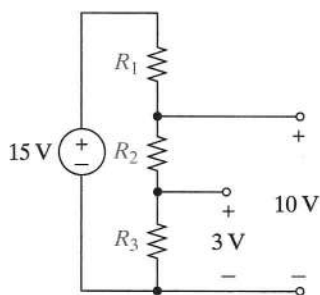


FIGURE P2-48

- 2-49 Select a value of R_x in Figure P2-49 so that $v_L = 3$ V.

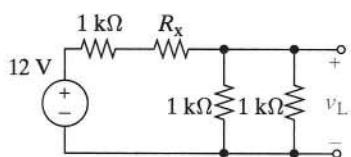


FIGURE P2-49

- 2-50 Select a value of R_x in Figure P2-50 so that $v_L = 2$ V. Repeat for 4 V and 6 V. *Caution:* R_x must be positive.

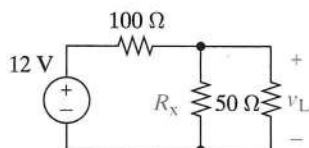


FIGURE P2-50

OBJECTIVE 2-6 CIRCUIT REDUCTION (SECT. 2-6)

Given a circuit consisting of linear resistors and an independent source, find selected signal variables using successive application of series/parallel equivalence, source transformations, and voltage/current division.

See Examples 2-20, 2-21, 2-22, 2-23 and Exercises 2-13, 2-14, 2-15

- 2-51 Use circuit reduction to find v_x and i_x in Figure P2-51.

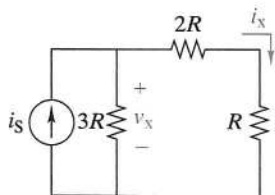


FIGURE P2-51

- 2-52 Use circuit reduction to find v_x and i_x in Figure P2-52.

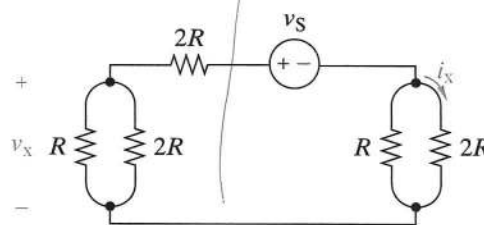


FIGURE P2-52

- 2-53 Use circuit reduction to find v_x and the power provided by the source in Figure P2-53.

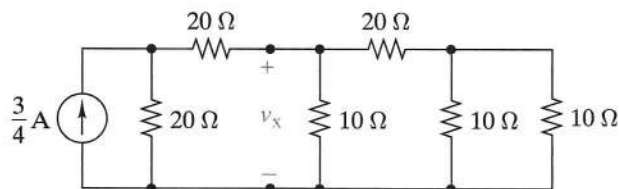


FIGURE P2-53

- 2-54 Use circuit reduction to find v_x and i_x in Figure P2-54.

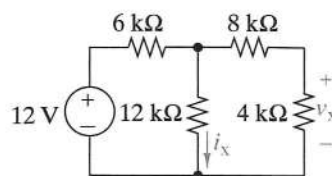


FIGURE P2-54

- 2-55 Use source transformation to find i_x in Figure P2-55.

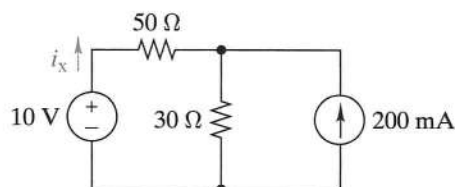


FIGURE P2-55

- 2-56 Select a value for R_x so that $i_x = 0$ A in Figure P2-56.

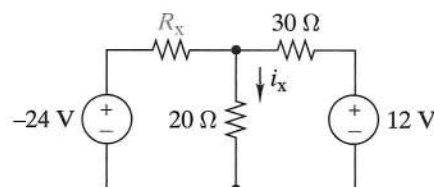


FIGURE P2-56

- 2-57 Use source transformations in Figure P2-57 to relate v_O to v_1 , v_2 , and v_3 .

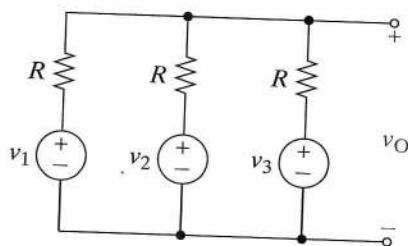


FIGURE P2-57

- 2-58 The current through R_L in Figure P2-58 is 40 mA. Use source transformations to find R_L .

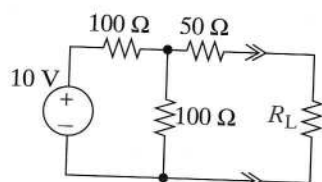


FIGURE P2-58

- 2-59 Select R_x so that 25 V are across it in Figure P2-59.

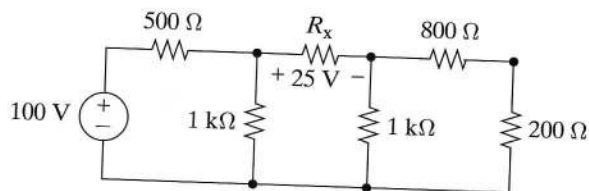


FIGURE P2-59

- 2-60 The box in the circuit in Figure P2-60 is a resistor whose value can be anywhere between 8 kΩ and 80 kΩ. Use circuit reduction to find the range of values of v_x .

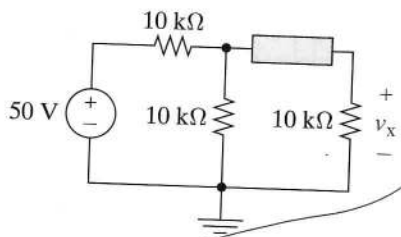


FIGURE P2-60

OBJECTIVE 2-7 COMPUTER-AIDED CIRCUIT ANALYSIS (SECT. 2-7)

Given an appropriate linear circuit, use circuit simulation and/or computational software to solve for the desired response. See Examples 2-24, 2-25 and Appendix B

- 2-61 A circuit is found to have the following element and connection equations:

$$\begin{aligned} v_1 &= 24 \text{ V} & -v_1 + v_2 + v_3 &= 0 \\ v_2 &= 8k i_2 & -v_3 + v_4 + v_5 &= 0 \\ v_3 &= 5k i_3 & i_1 + i_2 &= 0 \\ v_4 &= 4k i_4 & -i_2 + i_3 + i_4 &= 0 \\ v_5 &= 16k i_5 & -i_4 + i_5 &= 0 \end{aligned}$$

Use MATLAB to solve for all of the unknown voltages and currents associated with this circuit. Sketch one possible schematic that matches the given equations.

- 2-62 Consider the circuit of Figure P2-62. Use MATLAB to find all of the voltages and currents in the circuit.

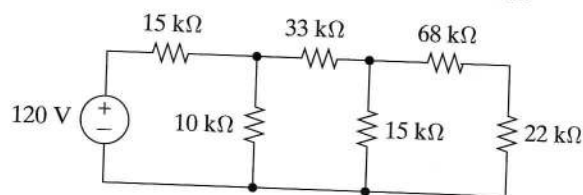


FIGURE P2-62

- 2-63 Consider the circuit of Figure P2-62 again. Use OrCAD to find all of the voltages and currents in the circuit.

- 2-64 Use OrCAD to find all the power dissipated or provided in the circuit of Figure P2-62. Verify that the total power in the circuit is zero.

INTEGRATING PROBLEMS

2-65 Nonlinear Device Characteristics

The circuit in Figure P2-65 is a parallel combination of a 50-Ω linear resistor and a varistor whose i - v characteristic is $i_v = 2.6 \times 10^{-5} v^3$. For a small voltage, the varistor current is quite small compared to the resistor current. For large voltages, the varistor dominates because its current increases more rapidly with voltage.

- Plot the i - v characteristic of the parallel combination.
- State whether the parallel combination is linear or nonlinear, active or passive, and bilateral or nonbilateral.
- Find the range of voltages over which the resistor current is at least 10 times as large as the varistor current.
- Find the range of voltages over which the varistor current is at least 10 times as large as the resistor current.

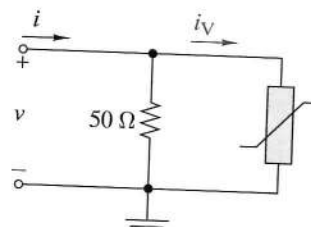


FIGURE P2-65

- 3-30(a)** Use the superposition principle to find v_O in terms of v_1 , v_2 , and R in Figure P3-30.
(b) Use MATLAB and node-voltage analysis to verify your answer symbolically.

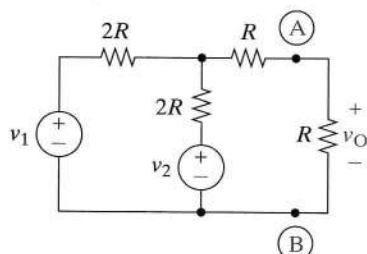


FIGURE P3-30

- 3-31(a)** Use the superposition principle to find v_O in terms of v_S , i_S , and R in Figure P3-31.
(b) Use MATLAB and node-voltage analysis to verify your answer symbolically.

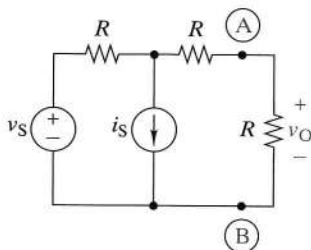


FIGURE P3-31

- 3-32** A linear circuit containing two sources drives a $100\text{-}\Omega$ load resistor. Source number 1 delivers 250 mW to the load when source number 2 is off. Source number 2 delivers 4 W to the load when source number 1 is off. Find the power delivered to the load when both sources are on. (Hint: The answer is not 4.25 W . Why?)

- 3-33** A linear circuit is driven by an independent voltage source $v_S = 10\text{ V}$ and an independent current source $i_S = 10\text{ mA}$. The output voltage is $v_O = 2\text{ V}$ when the voltage source is on and the current source off. The output is $v_O = 1\text{ V}$ when both sources are on. Find the output voltage when $v_S = 20\text{ V}$ and $i_S = -20\text{ mA}$.

- 3-34** A certain linear circuit has four input voltages and one output voltage v_O . The following table lists the output for different values of the four inputs. Find the input-output relationship for the circuit. Specifically, find an expression for v_O in terms of the four input voltages.

$v_{S1}\text{ (V)}$	$v_{S2}\text{ (V)}$	$v_{S3}\text{ (V)}$	$v_{S4}\text{ (V)}$	$v_O\text{ (V)}$
2	4	-4	1	20
1	2	2	1.5	-4
1	4	2	2	-1
0	5	3	-1	3

OBJECTIVE 3-3 THÉVENIN AND NORTON EQUIVALENT CIRCUITS (SECT. 3-4)

Given a circuit containing linear resistors and independent sources,

- (a) Find the Thévenin or Norton equivalent at a specified pair of terminals.
 (b) Use the Thévenin or Norton equivalent to find the signal delivered to linear or nonlinear loads.

See Examples 3-13, 3-14, 3-15, 3-16 and Exercises 3-23, 3-24, 3-25, 3-26

- 3-35(a)** Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-35.

- (b)** Use the equivalent circuit found in part (a) to find i_L in terms of $R_L = 47\text{ k}\Omega$.

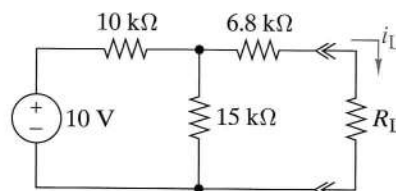


FIGURE P3-35

- 3-36(a)** Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-36.

- (b)** Use the equivalent circuit found in part (a) to find i_L in terms of i_S , R_1 , R_2 , and R_L .

- (c)** Check your answer to part (b) using current division.

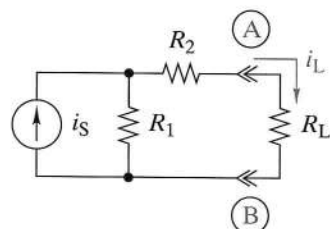


FIGURE P3-36

- 3-37** Find the Thévenin equivalent circuit seen by R_L in Figure P3-37. Find the voltage across the load when $R_L = 5\text{ }\Omega$, $10\text{ }\Omega$, and $50\text{ }\Omega$.

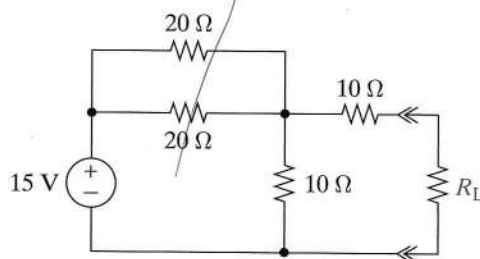


FIGURE P3-37

- 3-38 Find the Norton equivalent seen by R_L in Figure P3-38. Find the current through the load for $R_L = 4.7 \text{ k}\Omega$, $15 \text{ k}\Omega$, and $68 \text{ k}\Omega$.

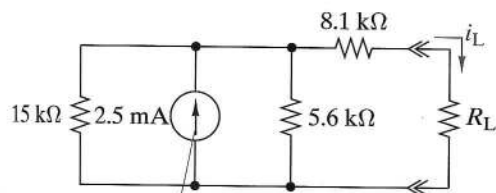


FIGURE P3-38

- 3-39 Find the Thévenin equivalent seen by R_L in Figure P3-39. Find the power delivered to the load when $R_L = 50 \text{ k}\Omega$ and $200 \text{ k}\Omega$.

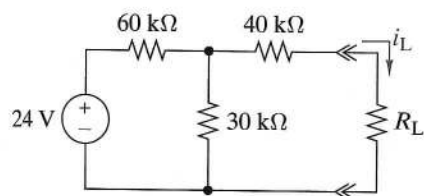


FIGURE P3-39

- 3-40 (a) Use OrCAD to find the Norton equivalent at terminals A and B in Figure P3-40. (Hint: Find the open-circuit voltage and short-circuit current at the requisite terminals.)
 (b) Use the Norton equivalent circuit found in part (a) to determine the power dissipated in R_L when it is equal to $37 \text{ k}\Omega$.
 (c) Use OrCAD to simulate both the original and the Norton equivalent circuits with $R_L = 37 \text{ k}\Omega$. Verify that the power dissipated by the load is the same in both situations.

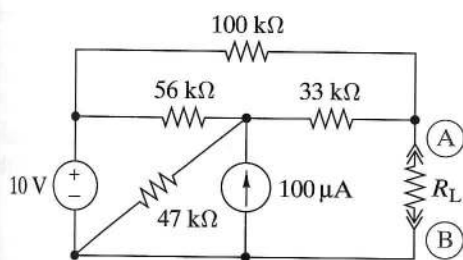


FIGURE P3-40

- 3-41 The purpose of this problem is to use Thévenin equivalent circuits to find the voltage v_x in Figure P3-41. Find the Thévenin equivalent circuit to the left of terminals A and B.

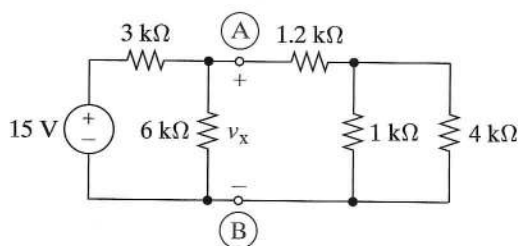


FIGURE P3-41

- 3-42 Assume that Figure P3-42 represents a model of the auxiliary output port of a car. The output current is $i = 1 \text{ A}$ when $v = 0 \text{ V}$. The output voltage is $v = 12 \text{ V}$ when a $20\text{-}\Omega$ resistor is connected between the terminals. Suppose you wanted to charge a 12-V battery by connecting the battery at the port. How much current would the port deliver to the battery?

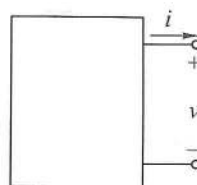


FIGURE P3-42

- 3-43 The i - v characteristic of the active circuit in Figure P3-42 is $5v + 500i = 100$. Find the output voltage when a $1\text{-k}\Omega$ resistive load is connected.
- 3-44 You have successfully completed Circuits I, and as part of an undergraduate work-study program your former professor has asked you to help her grade a Circuits I quiz. On the quiz, students were asked to find the power supplied by the source both to the $10\text{-k}\Omega$ load (R_L) and to the entire circuit as shown in Figure P3-44. Your professor asks you to help her by creating a grading sheet.
- (a) Solve the quiz and establish reasonable A, B, C, D, and F cuts for incorrect solutions.
- (b) A particular student correctly finds the Thévenin equivalent circuit seen by the resistive load and calculates the power to the load using V_T^2/R_L . He then does a source transformation, correctly finding the Norton equivalent of the circuit. He calculates the source power using $V_T \times I_N$. What grade would you give him?
- (c) Another student finds $P_L = 5.625 \text{ mW}$ and $P_S = -22.5 \text{ mW}$, but provides no work to justify her answers. What grade would you give her?
- (d) A third student first finds the Norton equivalent, and then finds the current through the load using a current